

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

COMBINED SCIENCE

4003/2

PAPER 2 Theory

SPECIMEN PAPER N2020

2 hours

Additional materials:

Answer sheets

Calculator (Optional)

String

Graph paper (as per candidate's request)

The Periodic Table is provided on page 15.

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top.
Check if the booklet has all the pages and ask the invigilator for a replacement if it has missing pages.

Section A

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

Section B

Answer any **two** questions.

Write your answers on the separate answer sheets provided.

Section C

Answer any **two** questions.

Write your answers on the separate answer sheets provided.

Section D

Answer any **two** questions.

Write your answers on the separate answer sheets provided.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question.

This question paper consists of 16 printed pages.
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Section A

*Answer **all** the questions in the spaces provided.*

1. (a) State any **two** differences between inhaled and exhaled air.

1

2

[2]

- (b) Transpiration is the loss of water through plant leaves.

- (i) State any **one** advantage of transpiration to the plant.

.....

.....

[1]

- (ii) State **one** disadvantage of excessive transpiration.

.....

.....

[1]

- (iii) State any **two** factors which increase the rate of transpiration.

1

2

[2]

2. (a) Describe a natural ecosystem.

.....
.....
.....

[2]

- (b) (i) Define the term *balanced diet*.

.....
.....
.....

[2]

- (ii) Describe the importance of calcium to a pregnant woman.

.....
.....
.....

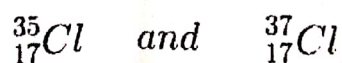
[2]

- (iii) State the advantage of eating liver.

.....
.....
.....

[1]

3. (a) Chlorine gas has two types of atoms as shown:



- (i) State the name given to the two types of the chlorine atoms.

.....

[1]

- (ii) Calculate the number of neutrons in ${}_{17}^{35}\text{Cl}$.

.....

[1]

- (b) Chlorine reacts with sodium to form sodium chloride, NaCl .

- (i) Name the type of bonding in sodium chloride.

[1]

- (ii) Draw a dot and cross diagram to show the bonding in sodium chloride.

[2]

- c) State any **two** physical properties of sodium chloride.

1
2

[2]

4. (a) Indigestion is caused by too much dilute hydrochloric acid in the stomach. It is cured by ingesting anti-acid tablets.

State, with a reason, the acid-base nature of the chemical present in the anti-acid tablets.

acid-base nature
reason
.....

[2]

- (b) (i) Iron is extracted from an iron compound found in haematite.
Name the iron compound in haematite.

.....

[1]

- (c) Two other solid raw materials are fed into the blast furnace together with haematite. Name the **two** raw materials and state a function for each of these materials.

raw material

function

raw material

function

[4]

5. (a) Fig.5.1 shows a stroke in the operation of an engine.

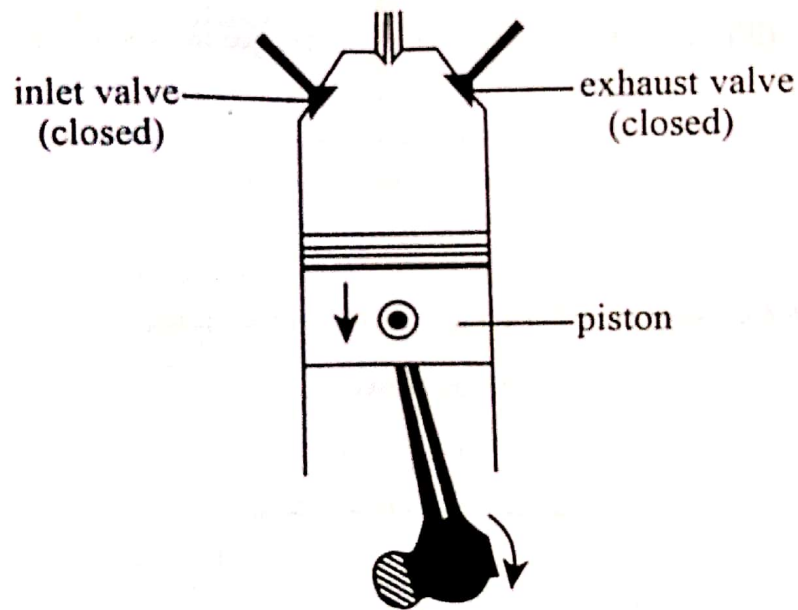


Fig.5.1

- (i) Identify, giving **two** reasons, the stroke shown.

stroke

reasons: 1

2

[3]

- (b) State the role of a fuel injector in a petrol engine.

.....
.....
.....

[1]

- (ii) State the role of a carburettor.

.....
.....

[1]

- (iii) Explain the advantage of a fuel injector over a carburettor.

.....
.....
.....

[2]

6. Fig.6.1 shows part of the design of a solar water heater.

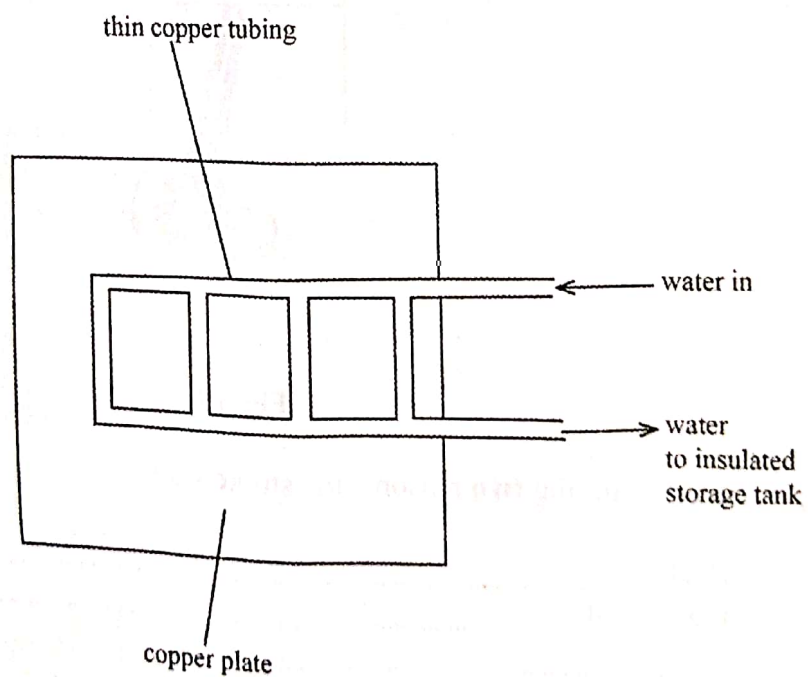


Fig.6.1

- (a) State, with a reason, the most suitable place for placing the solar water heater for best results.

place

reason

.....

[2]

- (b) State, with a reason, the paint colour on the copper plate.

colour

reason

[2]

- (c) Explain why

(i) a thin copper tubing is used,

(ii) the storage tank is insulated.

.....

[2]

Section B

Answer any **two** questions on the separate answer sheets provided.

7. (a) Fig.7.1 shows a sketch diagram to represent double circulation in mammals.

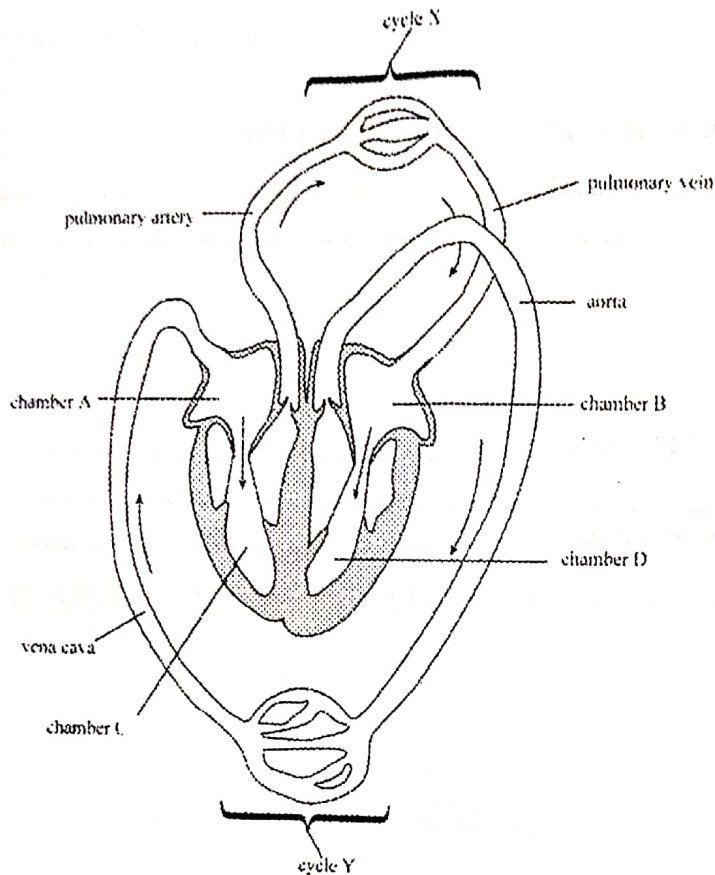


Fig.7.1

- (i) Deduce the types of circulation represented by cycles X and Y. [2]
- (ii) Suggest the reason for differences in the thickness of the walls of chambers C and D. [2]
- (iii) State any **three** symptoms of malaria. [3]
- (iv) State a symptom of ebola which is different from symptoms of malaria. [1]
- (v) State any **two** effects of inhaling glue. [2]

8. (a) Fig.8.1 shows a child suffering from a deficiency disease.

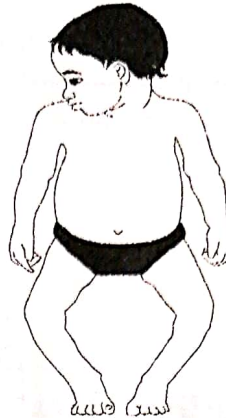


Fig.8.1

- (i) Name the deficiency disease which the child is suffering from. [1]
- (ii) Describe how the disease named in (i) could be prevented. [2]
- (b) Describe the route of the sperm from the testis to the oviduct. [4]
- (c) State **one** advantage of using condoms during sexual intercourse. [1]
- (d) Define the term *fertilisation*. [2]

9. (a) Fig.9.1 shows gaseous exchange in the alveolus of a mammal.

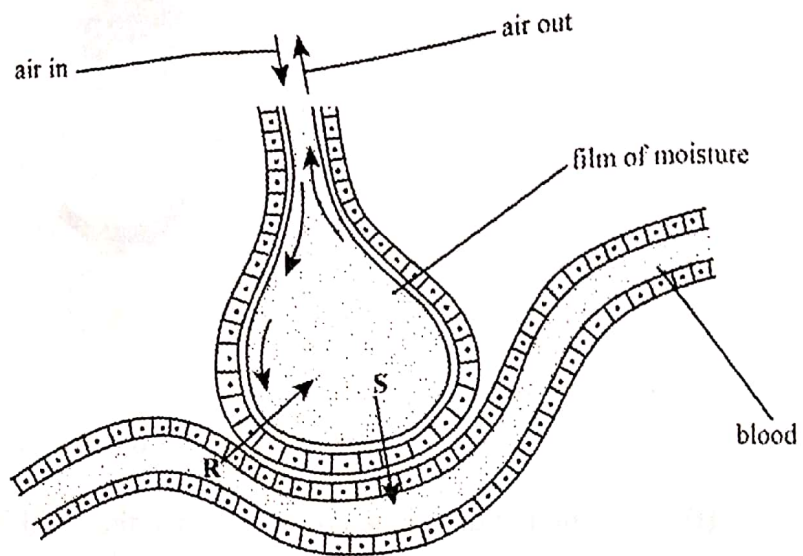


Fig.9.1

- (i) Name the gases moving in the directions shown by the arrows **R** and **S**. [2]
- (ii) Describe and explain how the alveolus is adapted for gaseous exchange. [4]
- (b) Define the terms *plasmolysis* and *turgidity*. 4]

Section C

Answer any two questions on the separate answer sheets provided.

10. (a) (i) Define the term *atom*. [1]
- (ii) State the **two** sub-atomic particles found in the nucleus of an atom. [2]
- (b) Determine the empirical formula of a compound made up of 75% by mass carbon and 25% by mass hydrogen. [4]
- (c) Sodium hydroxide solution reacts with dilute nitric acid to form a salt and water.
- (i) State the type of reaction that occurs. [1]
- (ii) Determine the chemical formula of the salt. [2]
11. (a) Outline the stages involved in the extraction of nitrogen from liquid air. [4]
- (b) Oxygen can be obtained from the electrolysis of acidified water.
- (i) Name the acid used to acidify the water. [1]
- (ii) Explain why the water is acidified. [2]
- (iii) Explain why the volume of oxygen obtained during the electrolysis process is half that of hydrogen. [2]
- (iv) State any **one** use of oxygen. [1]

12. Fig.12.1 shows the production of sulphuric acid by the contact process.

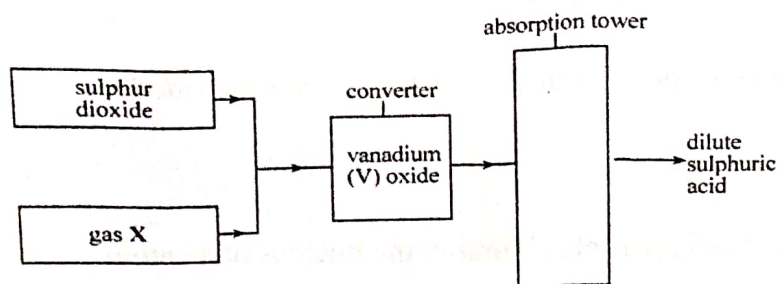


Fig.12.1

- (a) (i) Name gas X. [1]
- (ii) State the role of vanadium (V) oxide. [1]
- (iii) Explain why sulphur trioxide is **not** directly added to water. [2]
- (iv) Define the terms *exothermic* and *reversible*. [2]
- (v) Name the substance which is formed in the absorption tower. [1]
- (b) Ammonium sulphate, $(\text{NH}_4)_2\text{SO}_4$, is a fertilizer produced from sulphuric acid. Calculate the percentage composition by mass of nitrogen in ammonium sulphate. [3]

Section D

Answer any **two** questions on the separate answer sheets provided.

13. (a) Fig.13.1 shows an alternating current (a.c) generator.

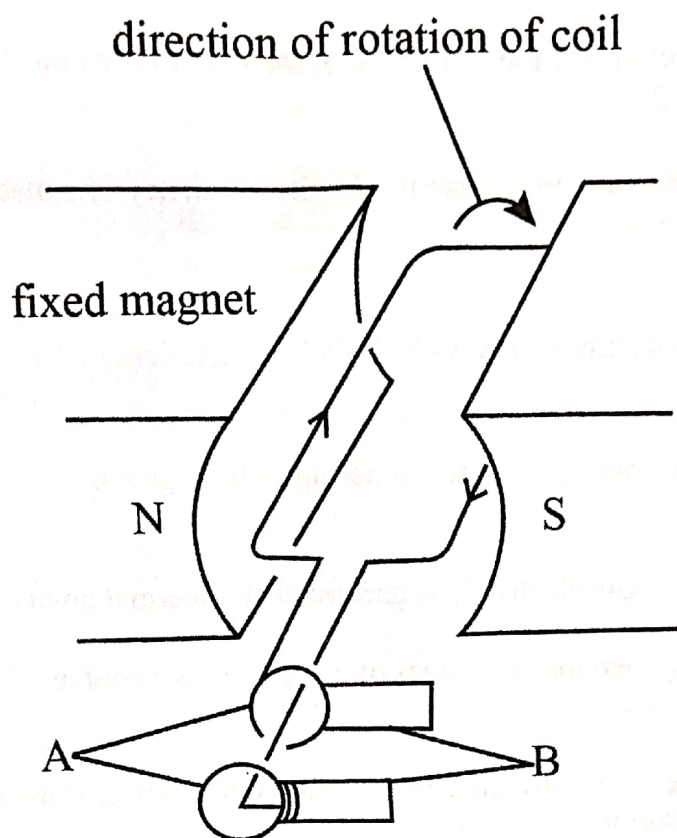


Fig.13.1

- (i) Name the parts labelled A and B. [2]
 - (ii) Describe how the a.c. generator produces electricity. [4]
 - (iii) Sketch a graph of output voltage of the generator against time. [2]
- (b) Explain the effect of using stronger magnets on the magnitude of the output voltage. [2]

14. (a) A gear system has ten teeth in the driving gear and thirty teeth in the driven gear.
- (i) Calculate the velocity ratio, VR, of the gear system. [2]
 - (ii) Determine the efficiency of the system if its mechanical advantage, MA, is 2. [2]
 - (iii) Give any **two** reasons why the efficiency of a machine is always less than 100 %. [2]
 - (iv) State any **two** ways by which the efficiency of a machine can be improved. [2]
- (b) State any **two** types of machines apart from gears. [2]
15. (a) Describe how electricity is generated at a thermal power station. [4]
- (b) State any **two** disadvantages of using coal as a source of fuel for a thermal power station. [2]
 - (c) Give the main difference between a thermal power station and a hydroelectric power station. [2]
 - (d) State the type of energy possessed by water which is in a dam. [1]
 - (e) State the Standard International (S.I) unit of energy. [1]

DATA SHEET

The Periodic Table of the Elements

Group																	
I	II											III	IV	V	VI	VII	O
		1 H Hydrogen															
3 Li Lithium	4 Be Beryllium											5 B Boron	6 C Carbon	7 N Nitrogen	8 O Oxygen	9 F Fluorine	10 Ne Neon
11 Na Sodium	12 Mg Magnesium											13 Al Aluminium	14 Si Silicon	15 P Phosphorus	16 S Sulphur	17 Cl Chlorine	18 Ar Argon
19 K Potassium	20 Ca Calcium	21 Sc Scandium	22 Ti Titanium	23 V Vanadium	24 Cr Chromium	25 Mn Manganese	26 Fe Iron	27 Co Cobalt	28 Ni Nickel	29 Cu Copper	30 Zn Zinc	31 Ga Gallium	32 Ge Germanium	33 As Arsenic	34 Se Selenium	35 Br Bromine	36 Kr Krypton
37 Rb Rubidium	38 Sr Strontium	39 Y Yttrium	40 Zr Zirconium	41 Nb Niobium	42 Mo Molybdenum	43 Tc Technetium	44 Ru Ruthenium	45 Rh Rhodium	46 Pd Palladium	47 Ag Silver	48 Cd Cadmium	49 In Indium	50 Sn Tin	51 Sb Antimony	52 Te Tellurium	53 I Iodine	54 Xe Xenon
55 Cs Cesium	56 Ba Barium	57 La Lanthanum	58 Ce Cerium	59 Pr Praseodymium	60 Nd Neodymium	61 Pm Promethium	62 Sm Samarium	63 Eu Europium	64 Gd Gadolinium	65 Tb Terbium	66 Dy Dysprosium	67 Ho Holmium	68 Er Erbium	69 Tm Thulium	70 Yb Ytterbium	71 Lu Lutetium	72 Hf Hafnium
87 Fr Francium	88 Ra Radium	89 Ac Actinium	90 Th Thorium	91 Pa Protactinium	92 U Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nihonium	103 Lr Lawrencium	104 Rn Radon

*58-71 Lanthanoid series
†90-103 Actinoid series

Key

a	X	b
a = relative atomic mass	X = atomic symbol	b = proton (atomic) Number

The volume of one mole of any gas is 28 dm³ at room temperature and pressure (r.t.p.)